

## Lecture Number Ten

MR. GRAHAM: Ladies and gentlemen, this is the last of our series of lectures. I hope that you will have found it as enjoyable and stimulating to listen to them as I have found it in preparing them.

The final talk is going to be something of departure, for it will address itself to speculation -- speculation in relation to security analysis.

Speculation, I imagine, is a theme almost as popular as love; but in both cases most of the comments made are rather trite and not particularly helpful. (Laughter.)

In discussing speculation in the context of this lecture it will be my effort to bring out some of the less obvious aspects of this important element in finance and in your own work.

There are three main points that I would like to make in this hour. The first is that speculative elements are of some importance in nearly all the work of the security analyst, and of considerable importance in part of his work; and that the over-all weight and significance of speculation has been growing over the past thirty years.

The second point is that there is a real difference between intelligent and unintelligent speculation, and that the methods of security analysis may often be of value in distinguishing between the two kinds of speculation.

My third point is that, despite the two foregoing statements, I believe that the present attitude of security analysts toward speculation is in the main unsound and unwholesome. The basic reason therefore is that our emphasis tends to be placed on the rewards of successful speculation rather than on our capacity to speculate successfully.

There is a great need, consequently, for a careful self-examining critique of the security analyst as speculator, and that means in turn a self-critique by the so-called typical investor, acting as speculator.

First, what do we mean by speculation? There is a chapter in our book on Security Analysis which is devoted to the distinctions between investment and speculation. I don't wish to repeat that material beyond recalling to you our concluding definition, which reads as follows:

½An investment operation is one which, on thorough analysis, promises safety of principal and a satisfactory return. Operations not meeting these requirements are speculative.½

That is a very brief reference to speculation. We could amplify it a bit by saying that in speculative operations a successful result cannot be predicated on the processes of security analysis. That doesn't mean that speculation can't be successful, but it simply means you can't be a successful speculator in individual cases merely by following our methods of security analysis.

Speculative operations are all concerned with changes in price. In some cases the emphasis is on price changes alone, and in other cases the emphasis is on changes in value which are expected to give rise to changes in price. I think that is a rather important classification of speculative operations. It is easy to give examples.

If at the beginning of 1946 a person bought U.S. Steel at around 80, chiefly because he believed that in the

latter part of bull markets the steel stocks tend to have a substantial move, that would clearly be a speculative operation grounded primarily on an opinion as to price changes, and without any particular reference to value.

On the other hand, a person who bought Standard Gas and Electric, four dollars preferred, sometime in 1945, at a low price, -- say at four dollars a share -- because he thought the plan which provided for its extinction was likely to be changed, was speculating undoubtedly. But there his motive was related to an analysis of value -- or rather to an expected change of value -- which, as it happened, was realized spectacularly in the case of the Standard Gas and Electric Preferred issue.

I think it is clear to you that in a converse sense nearly all security operations which are based essentially on expected changes, whether they are of price or of value, must be regarded as speculative, and distinguished from investment.

In our chapter on speculation and investment we discussed the concept of the speculative component in a price. You remember we pointed out that a security might sell at a price which reflected in part its investment value and in part an element which should be called speculative.

The example we gave back in 1939-1940, with considerable trepidation, was that of General Electric. We intentionally picked out the highest-grade investment issue we could find to illustrate the element of speculation existing in it. Of the price of \$38, which it averaged in 1939, we said the analyst might conclude that about \$25 a share represented the investment component and as much as \$13 a share represented the speculative component. Hence in this very high-grade issue about one-third of the average price in a more or less average market represents a speculative appraisal. That example, which showed how considerable was the speculative component in investment securities, I think is pretty typical of security value developments since World War I. I believe it justifies and explains the first point that I wish to make, namely, that speculative elements have become more and more important in the work of the analyst. I think only people who have been in Wall Street for a great many years can appreciate the change in the status of investment common stocks that took place in the last generation, and the extent to which speculative considerations have obtruded themselves in all common stocks.

When I came down to the Street in 1914, an investment issue was not regarded as speculative, and it wasn't speculative. Its price was based primarily upon an established dividend. It fluctuated relatively little in ordinary years. And even in years of considerable market and business changes the price of investment issues did not go through very wide fluctuations. It was quite possible for the investor, if he wished, to disregard price changes completely, considering only the soundness and dependability of his dividend return, and let it go at that -- perhaps every now and then subjecting his issue to a prudent scrutiny.

That fact is illustrated on the blackboard by taking the rather extreme case of the Consolidated Gas Company, now Consolidated Edison Company, during the years of the first postwar boom and depression -- namely, 1919-1923. These vicissitudes really affected the company quite severely; for you will notice that its earnings suffered wide fluctuations, and got down in 1920 to only \$1.40 a share for the \$100 par value stock. Yet during that period it maintained its established dividend of seven dollars and its price fluctuation was comparatively small for a major market swing -- that is, it covered a range of 106 down to 71.

If we go back to the years 1936-1938, which in the textbooks is now referred to as a mere "recession" that lasted for a year, we find that Consolidated Edison Company, with no changes in earnings to speak of, had extraordinarily wide changes in price. During the year 1937 alone, it declined from about 50 to 21, and the following year went down to 17. During that period it actually raised its dividend, and its earnings were very stable. (See comparative data in the following table.)

The much wider fluctuations in investment common stocks that have come about since World War I have made it practically impossible for buyers of common stocks to disregard price changes. It would be extremely unwise -- and hypocritical -- for anybody to buy a list of common stocks and say that he was interested only in his dividend return and cared nothing at all about price changes.

The problem is not whether price changes should be disregarded -- because clearly they should not be -- but rather in what way can the investor and the security analyst deal intelligently with the price changes which take place.

I would like to go back for a moment to our statement that in the case of General Electric a considerable portion of the price in 1939 reflected a speculative component. That arises from the fact that investors have been willing to pay so much for so-called quality, and so much for so-called future prospects, on the average, that they have themselves introduced serious speculative elements into common stock valuations. These elements are bound to create fluctuations in their own attitude, because quality and prospects are psychological factors. The dividend, of course, is not a psychological factor; it is more or less of a fixed datum. Matters of the former kind -- I am speaking now of prospects and quality -- are subject to wide changes in the psychological attitude of the people who buy and sell stocks. Thus we find that General Electric will vary over a price range almost as wide as that of any secondary stock belonging in more or less the same price class.

Going ahead from 1939 to 1946, we find that General Electric declined from 44 1/2 down to 21 1/2 and came back again to 52 in 1946, and has since declined to 33, or thereabouts. These are wide fluctuations. I think they justify my statement that a very considerable part of the price of General Electric must be regarded as speculative and perhaps temporary.

I think also you might say that the pure investment valuation of \$25 for General Electric could be said to be justified by the sequel, since there were opportunities both in 1941 and 1942 to buy the stock at those levels. It is also true that the price movement of General Electric was not as favorable between 1939 and 1946 as that of other stocks, and I think that reflects the rather over-emphasized speculative element that appeared in General Electric before World War II.

Speculative components may enter into bonds and preferred stocks as well as into common stocks. But a high-grade bond, almost by definition, has practically no speculative component. In fact, if you thought it had a large speculative component, you would not buy it for investment nor would you call it high grade. But there is one important factor to be borne in mind here. A rise in interest rates may cause a substantial decline in the price of a very good bond. But even in that event a high-grade bond may be valued on its amortized basis throughout the period that it runs, and the price fluctuations could therefore be ignored by a conventional treatment of value. As most of you know, that is exactly what is done in the insurance company valuation methods which we were discussing recently. High-grade bonds are valued from year to year on an amortized basis, without reference to price fluctuations.

It may be a pleasant thing for the security analyst to get away from the speculative components that are found chiefly in common stocks and which are so troublesome, and to concentrate on the more responsive and more controllable elements in bond analysis. Wall Street, I believe, has improved very greatly its technique of bond analysis since 1929. But it is one of the ironies of life that just when you have got something really under control it is no longer as important as it used to be. I think we must all admit that bond analysis plays a very much smaller part in the work of the analyst and in the activities of the investor than it used to. The reason is perfectly obvious: The greater portion of bond investments now consist of U.S. government bonds, which do not require or lend themselves to a formal bond analysis.

While it is true that for the minor portion of corporate bonds that remain you can go through all the motions

of careful bond analysis, even that is likely to be somewhat frustrating. For I am sure that a really competent bond analyst is almost certain to come up with the conclusion in nearly every case that the typical buyer would be better off with a Government bond than with a well-entrenched corporate security. The purchase of these corporate securities in the present market is a kind of pro forma affair by the large institutions who, for semi-political reasons, desire to have corporate bonds in their portfolios as well as Government bonds. The result is that the wide field of bond analysis, which used to be so important to and so rewarding to the bond investor, must now, I think, be written down pretty far in terms of practical interest.

So much, then, for my first point: That willy-nilly we security analysts find that more and more significance attaches to speculative elements in the securities that we are turning our attention to.

On the second point, which relates to the analyst's role in distinguishing intelligent from unintelligent speculation, I would like to treat that matter chiefly by some examples. I have picked out four low-price securities, which I think would illustrate the different kinds of results which an analyst may get from dealing with primarily speculative securities. These are, on the one hand, Allegheny Corp. Common, which sold at the end of the month at five, and Graham-Paige Common, which sold at five; and, on the other hand, General Shareholdings, which sold at four, and Electric Bond and Share six dollars Preferred  $\frac{1}{2}$  Sub  $\frac{1}{2}$ , which could be bought yesterday at the equivalent of three.

When we first look at these securities, they all seem pretty much the same -- namely, four speculative issues, which they certainly are. But a deeper examination by a security analyst would reveal a quite different picture in the two pairs of cases.

In the case of General Shareholdings we have the following: This is the common stock of an investment company, which has \$21.5-million of total assets, with senior claims of \$12-million, and a balance of about \$9.5-million for the common. The common is selling for \$6,400,000 in the market. That means that in General Shareholdings you have both a market discount from the apparent present value of the stock and an opportunity to participate in a highly leveraged situation. For if you pay \$6.4-million of the gross asset value; and consequently every ten per cent of increase in total asset value would mean a 30 per cent increase in the book value of the common.

Furthermore, you are practically immune from any danger of serious corporate trouble; because the greater portion of the senior securities -- in fact, five-sixths of it -- is represented by a preferred stock on which dividends do not have to be paid and on which there is no maturity date.

Consequently, in the General Shareholdings case, you have that typically attractive speculative combination of (a) a low-price  $\frac{1}{2}$  ticket of entry  $\frac{1}{2}$  into a fairly large situation; and (b) instead of paying more than the mathematical value of your ticket, you are paying less; and  $\frac{1}{2}$  if you assume that wide fluctuations are likely to occur in both directions over the years, you stand to gain more than you can lose from these fluctuations.

So much for General Shareholdings, viewed analytically.

By contrast, if you go to Allegheny Corporation at five, although it seems at first to be a somewhat similar situation -- namely an interest in an investment company portfolio -- you find the mathematical picture completely different. At the end of 1945 the company had about \$85-million of assets, and against it there were \$125-million claims in the form of bonds and preferred stocks, including unpaid dividends. Thus the common stock was about \$40-million  $\frac{1}{2}$  under water.  $\frac{1}{2}$  Yet at five you would be paying \$22-million for your right to participate in any improved value for the \$85-million of assets, -- after the prior claims were satisfied.

The security analyst would say that there is plenty of leverage in that situation, of course; but you are paying

so much for it, and you are so far removed from an actual realizable profit, that it would be an unintelligent speculation.

The fact of the matter is you would need a 70 per cent increase in the value of the Allegheny portfolio merely to be even with the market price of the common as far as asset value coverage is concerned. In the case of General Shareholdings, if you had a 70 per cent increase in the value of its portfolio, you would have an asset value of about \$15 a share for the common, as against a market price of around four.

Thus, from the analytical standpoint, while Allegheny and General Shareholdings represent approximately the same general picture, there is a very wide quantitative disparity between the two. One turns out to be an intelligent and the other an unintelligent speculation.

Passing now to Graham-Paige at five dollars, we find another type of situation. Here the public is paying about \$24-million for a common stock which represents about \$8-million of asset value, most of which is in Kaiser-Fraser stock. This you can buy if you want in the open market, instead of having to pay three times as much for it. The rest of the price represents an interest in \$3-million of assets in the farm equipment business -- which may prove profitable, as any business may be profitable. The only weakness to that is that there is no record of profitable operations here, and you are paying a great many millions of dollars merely for some possibilities. That, in turn, would be regarded as an unintelligent speculation by the security analyst.

Let us move on now to the Electric Bond and Share Stubs, which I shall describe briefly. They represent what you would have left if you had bought Electric Bond and Share Preferred at \$73 yesterday and had then received \$70 a share that is now to be distributed. What remains is an interest in a possible ten dollar payment, your claim to which is to be adjudicated by the SEC and the courts. That ten dollars represents the premium above par to which Electric Bond and Share Preferred would be entitled if it were called for redemption. The question to be decided is whether the call price, the par value, or some figure in between should govern in this case.

It should be obvious, I think, that that is a speculative situation. You may get ten dollars a share out of it for your three dollars, and you may get nothing at all, or you may get something in between. But it is not a speculative operation that eludes the techniques of the security analyst. He has means of examining into the merits of the case and forming an opinion based upon his skill, his experience, and the analogies which he can find in other public utility dissolutions.

If we were to assume that the Electric Bond and Share Stubs have a 50-50 chance of getting the ten dollar premium, then he would conclude that at three dollars a share they are an intelligent speculation. For the mathematics indicates that, in several such operations, you would make more than you would lose in the aggregate. These examples lead us, therefore, to what I would call a mathematical or statistical formulation of the relationship between intelligent speculation investment. The two, actually, are rather closely allied.

Intelligent speculation presupposes at least that the mathematical possibilities are not against the speculation, basing the measurement of these odds on experience and the careful weighing of relevant facts.

This would apply for example, to the purchase of common stocks at anywhere within the range of value that we find by our appraisal method. If you go back for a moment to our appraisal of American Radiator, you may recall that in our fifth lecture we went through a lot of calculations and came out with the conclusion that American Radiator was apparently worth between \$15 and \$18 a share. If we assume that that job was well done, we could draw these conclusions. The investment value of American Radiator is about \$15; between 15 and 18 you would be embarking on what might be called an intelligent speculation, because it would be justified by your appraisal of the speculative factors in the case. If you went beyond the top range of \$18 you would be going over into the field of unintelligent speculation.

If the probabilities, as measure by our mathematical test, are definitely in favor of the speculation, then we can transform these separate intelligent speculations into investment by the simple device of diversification. That, I think, is a clue to the most successful and rewarding treatment of speculation in Wall Street. The idea, in fine, is simply to get the odds on your side by processes of skillful, experienced calculation.

Going back to our Electric Bond and Share example, if we really are skillful in our evaluation of the possibilities here, and reach this conclusion of a 50-50 possibility, then we could consider Electric Bond and Share Stubs as part of an investment operation consisting of, say, ten such ventures of a diversified character. For in ten such operations you would get \$50 back for an investment of \$30, if you have average luck. That is, you would get ten dollars each on five of them and you would get nothing on another five, and your aggregate return would be \$50.

Very little has been done in Wall Street to work out these arithmetical aspects of intelligent speculation based on favorable odds. In fact, the very language may be strange to most of you. Yet it oughtn't to be. If we are allowed to commit some misdemeanor by making some mild comparisons between Wall Street and horse-racing, the thought might occur to some of us that the intelligent operator in Wall Street would try to follow the technique of the bookmaker rather than the technique of the man who bets on the horses. Further, if we assume that a very considerable amount of Wall Street activity must inevitably have elements of chance in it, then the sound idea would be to measure these chances as accurately as you can, and play the game in the direction of having the odds on your side.

Therefore, quite seriously, I would recommend to this group, and to any other, that the mathematical odds of speculation in various types of Wall Street operations would provide a full and perhaps a profitable field of research for students.

Let us return for a moment to Allegheny Common and Graham-Paige Common, which we characterized as unintelligent speculation from the analyst's viewpoint. Is not this a dangerous kind of statement for us to make? Last year Graham-Paige sold as high as 16, and Allegheny as high as eight and one quarter, against the current figure of five. It must be at least conceivable that their purchase today might turn-out very well, either because (a) the abilities of Mr. Young or Mr. Fraser will create real value where none or little now exists, or (b) the stocks will have a good speculative  $\frac{1}{2}$  move,  $\frac{1}{2}$  regardless of value.

Both of these possibilities exist, and the analyst cannot afford to ignore them. Yet he may stick to his guns in characterizing both stocks as unintelligent speculations, because his experience teaches him that this type of speculation does not work out well on the average. One reason is that the people who buy this kind of stock at five are more likely to buy more at ten than to sell it. Consequently, they usually show losses in the end, even though there may have been a chance in the interim to sell out to even less intelligent buyers. Thus, in the end, the criterion of both intelligent and unintelligent speculation rests on the results of diversified experience.

When I come to my third point I am going to indicate how very different are the ordinary and customary attitudes toward speculative risk in Wall Street than those we have been discussing. But I think I ought to pause here for a minute, since I finished my second point, and see if there are some questions to be asked on this exposition.

QUESTION: By diversification, as in the case of Electric Bond and Share Stubs -- you wouldn't concentrate on ten situations similar in the way of redemption of preferred. You would want to diversify with Electric Bond and Share stocks and General Shareholdings, and some others; entirely different situations?

MR. GRAHAM: Yes, the approach is not based on the character of the operation, but only on the mathematical odds which you have been able to determine to your own satisfaction. It doesn't make any

difference what you are buying, whether a bond or a stock or in what field, if you are reasonably well satisfied that the odds are in your favor. They are all of equal attractiveness, and they all belong equally in your diversification. You make a further sound point, and that is that you are not really diversifying if you went into ten Electric Bond and Share situations -- all substantially the same. You would not really be diversifying, because that is practically the same thing as buying ten shares of Electric Bond and Share instead of buying one share of each; since the same factors would apply to all of them. That point is well taken. For real diversification; you must be sure that the factors that make for success or failure differ in one case from another.

\*\*\* QUESTION: As for that 50-50 chance, why didn't you come up with sixty-forty -- in Bond and Share? I don't see how you can be so mathematically precise.

MR. GRAHAM: Of course you are right in saying that, and I am glad you raised the point. This is not something that admits of a Euclidean demonstration. But you can reach the conclusion that the chances are considerably better than seven to three, let us say -- which are the odds that are involved in your purchase -- without being exactly sure whether they are 50-50 or sixty-forty. Broadly speaking, you simply say you think the chances are at least even in your favor, and you let it go at that. But that is enough for the purpose. You don't have to be any more accurate for practical action.

(Now, bear in mind I am not trying to imply here that the figure given is necessarily my conclusion as to what the odds in the Bond and Share are. Any of you are perfectly competent to study that situation and draw a conclusion based upon what has taken place in other utility redemptions. I am only using the Stubs for purposes of illustration. I should point out that the market does not seem to be very intelligent in paying the same price for the five dollar Preferred Stubs as for the six dollar Preferred Stubs.)

The final subject that I have is the current attitude of security analysts toward speculation. It seems to me that Wall Street analysts show an extraordinary combination of sophistication and naiveté in their attitude toward speculation. They recognize, and properly so, that speculation is an important part of their environment. We all know that if we follow the speculative crowd we are going to lose money in the long run. Yet, somehow or other, we find ourselves very often doing just that. It is extraordinary how frequently security analysts and the crowd are doing the same thing. In fact, I must say I can't remember any case in which they weren't. (Laughter.)

It reminds me of the story you all know of the oil man who went to Heaven and asked St. Peter to let him in. St. Peter said, 'Sorry, the oil men's area here is all filled up, as you can see by looking through the gate.' The man said, 'That's too bad, but do you mind if I just say four words to them?' And St. Peter said, 'Sure.' So the man shouts good and loud, 'Oil discovered in hell!' Whereupon all the oil men begin trooping out of Heaven and making a beeline for the nether regions. Then St. Peter said, 'That was an awfully good stunt. Now there's plenty of room, come right in.' The oil man scratches his head and says, 'I think I'll go with the rest of the boys. There may be some truth in that rumor after all.' (Laughter.)

I think that is the way we behave, very often, in the movements of the stock market. We know from experience that we are going to end up badly, but somehow there may be some truth in the rumor; so we go along with the boys.

For some reason or other, all security analysts in Wall Street are supposed to have an opinion on the future of the market. Many of our best analytical brains are constantly engaged in the effort to forecast the movement of prices. I don't want to fight out the battle over again here, as to whether their activity is sound or not. But I would like to make one observation on this subject.

The trouble with market forecasting is not that it is done by unintelligent and unskillful people. Quite to the contrary, the trouble is that it is done by so many really expert people that their efforts constantly neutralize each other, and end up almost exactly in zero.

The market already reflects, almost at every time, everything that the experts can reliably say about its future. Everything in addition which they say is therefore unreliable, and it tends to be right just about half the time. If people analyzing the market would engage in the proper kind of self-criticism, I am sure they would realize that they are chasing a will-o'-the-wisp.

Reading recently the biography of Balzac, I recalled that novel of his called, *The Search for the Absolute*, which some of you may have read. In it a very intelligent doctor spends all his time looking for something which would be wonderful if he found it, but which he never finds. The reward for being consistently right on the market is enormous, of course, and that is why we are all tempted. But I think you must agree with me that there is no sound basis for believing that anyone can be constantly right in forecasting the stock market. In my view it is a great logical and practical mistake for security analysts to waste their time on this pursuit.

Market forecasting, of course, is essentially the same as market timing. On that subject let me say that the only principle of timing that has ever worked well consistently is to buy common stocks at such times as they are cheap by analysis, and to sell them at such times as they are dear, or at least no longer cheap, by analysis.

That sounds like timing; but when you consider it you will see that it is not really timing at all but rather the purchase and sale of securities by the method of valuation. Essentially, it requires no opinion as to the future of the market; because if you buy securities cheap enough, your position is sound, even if the market should continue to go down. And if you sell the securities at a fairly high price you have done the smart thing, even if the market should continue to go up.

Therefore, at the conclusion of this course, I hope you will permit me to make as strong a plea as I can to you security analysts to divorce yourselves from stock market analysis. Don't try to combine the two -- security analysis and market analysis -- plausible as this effort appears to many of us; because the end-product of that combination is almost certain to be contradiction and confusion.

On the other hand, I should greatly welcome an effort by security analysts to deal intelligently with speculative operations. To my mind the prerequisite here is for the quantitative approach, which is based on the calculation of the probabilities in each case, and a conclusion that the odds are strongly in favor of the operation's success. It is not necessary that this calculation be completely dependable in each instance, and certainly not mathematically precise, but only that it be made with a fair degree of knowledge and skill. The law of averages will take care of minor errors and of the many individual disappointments which are inherent in speculation by its very definition.

It is a great mistake to believe that a speculation has been unwise if you lose money at it. That sounds like an obvious conclusion, but actually it is not true at all. A speculation is unwise only if it is made on insufficient study and by poor judgment. I recall to those of you who are bridge players the emphasis that the bridge experts place on playing a hand right rather than on playing it successfully. Because, as you know, if you play it right you are going to make money and if you play it wrong you lose money -- in the long run.

There is a beautiful little story, that I suppose most of you have heard, about the man who was the weaker bridge player of the husband-and-wife team. It seems he bid a grand slam, and at the end he said very triumphantly to his wife, "I saw you making faces at me all the time, but you notice I not only bid this grand slam but I made it. What can you say about that?" And his wife replied very dourly, "If you had played it right you would have lost it." (Laughter.)



There is a great deal of that in Wall Street, particularly in the field of speculation, when you are trying to do it by careful calculation. In some cases the thing will work out badly. But that is simply part of the game. If it was bound to work out rightly, it wouldn't be a speculation at all, and there wouldn't be the opportunities of profit that inhere in sound speculation. It seems to me that is axiomatic.

\*\*\* I know something of the practical problems that confront the security analyst who wants to act logically all the time, and who wants to confine himself only to that area of financial work in which he can say with confidence that his work and his conclusions are reasonably dependable. The analysts all complain to me that they can't do that because they are expected by their customers and their employers to do something else, to give them off-the-cuff speculative judgments and market opinions. One of these days I am sure the security analysts will divide themselves completely from the market analysts.

It would be very nice to have a two-year trial period in which the market analysts would keep track of what they have accomplished through the period and security analysts would keep track of what they have accomplished. I think it would be rather easy to tell in advance who would turn in the better score. That is really the pay-off. I think that eventually the employers and the customer will come to the conclusion that it is better to let the security analysts be security analysts -- which they know how to do -- and not other kinds of things, particularly market analysts, which they don't know how to do and they will never know how to do.

I would like to make some final observations, relating to a long period of time, as to what has happened to the conduct of business in Wall Street.

If you can throw your mind, as I can, as far back as 1914, you would be struck by some extraordinary differences in Wall Street then and today. In a great number of things, the improvement has been tremendous. The ethics of Wall Street are very much better. The sources of information are much greater, and the information itself is much more dependable. There have been many advances in the art of security analysis. In all those respects we are very far ahead of the past.

In one important respect we have made practically no progress at all, and that is in human nature. Regardless of all the apparatus and all the improvements in techniques, people still want to make money very fast. They still want to be on the right side of the market. And what is most important and most dangerous, we all want to get more out of Wall Street than we deserve for the work we put in.

There is one final area in which I think there has been a very definite retrogression in Wall Street thinking. That is in the distinctions between investment and speculation, which I spoke about at the beginning of this lecture. I am sure that back in 1914 the typical person had a much clearer idea of what he meant by investing his money, and what he meant by speculating with his money. He had no exaggerated ideas of what an investment operation should bring him, and nearly all the people who speculated knew approximately what kind of risks they were taking.